Some database maintenance is required after Direct Updates as well. Due to the design of PostgreSQL, an UPDATE statement for a row leaves around a former version of the row that's now considered "dead"; dead rows can also be called *bloat*. Database maintenance on a table removes bloat, attempting to reclaim the space consumed by dead rows. Direct Updates cause bloat and Table Copying doesn't.

For Direct Updates, you'll run the VACUUM and REINDEX maintenance commands after scrubbing.

A goal of a the Performance database scrubbing system is that it runs fast. With a fast running process, your team will have an easier time automating the steps and repeating the whole process periodically.

To avoid the slowdown caused by Direct Updates with large tables, you'll use a second technique called Table Copying for very large tables with millions of rows. Table Copying uses a trick to replace the original table with a copy of the table and then swap the names around.

Table Copying doesn't require post-copy database maintenance either.

Table Copying also runs faster by strategically deferring some things like constraint checks. Database constraint checks from DML operations like UPDATE statements normally run on updates and provide an important data consistency benefit. Table Copying temporarily removes the constraints and then adds them back later. This speeds up the process.

You've now seen a brief comparison of *Direct Updates* and *Table Copying* but haven't yet put these into practice. First you'll need a way to scrub the sensitive text.

Starting An Email Scrubber Function

For your first scrubber function you'll scrub the value of the users table email column. This column has a varchar type and the email addresses have a variety of domains like gmail.com. Your design goal is to preserve the email domain portion because it does not uniquely identify a person, but replace the first portion before the "@" symbol.

To do this you'll create a database function called scrub_email() using built-in SQL functions. As a Rails developer you may not have written database functions before. Don't worry, you'll be gradually introduced to it.

PostgreSQL functions can be written with SQL (LANGUAGE SQL). For more advanced needs, there is a built-in Procedural Language called *PL/pgSQL*.